Project Management System

Software Requirements Specification

Version 1.0

Team No. 11

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SRN	Name	Role
PES2UG24CS474	Shivansh Sharma	Team Lead
PES2UG24CS467	Shibi R Shetty	Member
PES2UG24CS466	Sheetal S Bevinakatti	Member
PES2UG24CS471	Shivanagouda I Patil	Member

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Revision History

Name	Date	Reason For Changes	Version
Team 11	05-Sep-2026	Initial draft of SRS	1.0

1. Introduction

1.1 Purpose

This document specifies the software requirements for the Project Management System (PMS), Version 1.0, being developed by Team 11 as a mini-project. It describes the functional and non-functional requirements for a web-based application that enables teams to plan, track, and manage projects through workflow states, Kanban boards, effort estimation, and collaboration features. This SRS covers the complete scope of the system as planned for the initial release.

1.2 Intended Audience and Reading Suggestions

This document is intended for the development team (for implementation guidance), the course instructor/evaluator (for assessing project scope and completeness), and any reviewers of the mini-project. Section 2 gives a high-level overview of the product; Section 3 covers external interfaces; Section 5 details each functional requirement; Section 6 covers non-functional requirements; the Appendices provide supporting reference material. Readers primarily interested in "what the system must do" should focus on Sections 2 and 5.

1.3 Product Scope

The Project Management System is a single web application that combines project workflow management, task/issue tracking, Kanban-style visualization, effort estimation, sprint/milestone planning, and reporting into one platform. Its goal is to give team members a simple way to create and move tasks through a workflow, and to give project managers real-time visibility into project progress, without needing separate tools for planning, tracking, and reporting. The system targets small to medium-sized software/project teams and is built using Python (Django) or a MERN (MongoDB, Express, React, Node.js) stack.

1.4 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications
- Project Management System Requirements document, Team 11 (source requirements list)
- Atlassian Jira and Trello documentation, referenced for Kanban/workflow conventions

2. Overall Description

2.1 Product Perspective

The PMS is a new, self-contained web product; it is not a follow-on to an existing in-house system. It draws inspiration from established tools such as Jira and Trello but is being built independently as a mini-project. The system consists of a browser-based client (React/JS front end or Django templates), a back-end application server exposing REST APIs, and a relational/NoSQL database for persistence. It integrates with external tools (Slack, GitHub, email) via its REST API for notifications and issue linking.

2.2 Product Functions

At a high level, the PMS allows users to:

- Register, authenticate, and operate under role-based permissions (Admin, Project Manager, Team Member)
- Create and manage projects and their team membership
- Create, edit, assign, and track tasks/issues on a drag-and-drop Kanban board
- Define custom workflows and valid state transitions per project
- Estimate effort (story points/time) and compare estimated vs. actual effort
- Plan and close sprints/milestones
- View dashboards with burndown/velocity charts and task distribution
- Receive notifications, comment on tasks, attach files, and search/filter across projects
- Generate downloadable reports and view an audit trail of changes

2.3 User Classes and Characteristics

User Class	Characteristics
Admin	Full system access; manages users, roles, and system-wide settings. Technical familiarity assumed to be moderate to high.
Project Manager	Creates and configures projects, workflows, and sprints; primary consumer of dashboards and reports. Moderate technical expertise.
Team Member	Creates, updates, and moves tasks assigned to them; comments and collaborates. Least technical expertise assumed; interface must be intuitive.

2.4 Operating Environment

The PMS is a web application accessed through modern browsers (Chrome, Firefox, Edge, Safari) on desktop and tablet devices. The back end runs on a Linux/Windows server environment using either Python/Django or Node.js (MERN stack), backed by a relational database (PostgreSQL/MySQL) or MongoDB. No native mobile application is planned for the initial release; the web UI is responsive for tablet use.

2.5 Design and Implementation Constraints

- Must be implemented using Python/Django or the MERN stack (or an equivalent approved by the course instructor).
- Must follow an MVC/MVT architectural pattern for maintainability.
- Must expose functionality through a documented REST API to support third-party integrations.

- Passwords must be hashed (e.g., bcrypt) and all traffic served over HTTPS.
- Development timeline is constrained by the academic semester schedule for the mini-project.

2.6 Assumptions and Dependencies

- Users have access to a modern web browser and a stable internet connection.
- Third-party integrations (Slack, GitHub, email) depend on the availability and API stability of those external services.
- The project assumes a small-to-moderate concurrent user base consistent with a mini-project deployment, not enterprise-scale load.

3. External Interface Requirements

3.1 User Interfaces

The system provides a responsive web UI with: a login/registration screen; a project dashboard; a Kanban board supporting drag-and-drop between columns (Backlog, To Do, In Progress, Review, Done); a task detail view with fields for title, description, priority, due date, labels, comments, and attachments; and reporting/analytics screens showing burndown and velocity charts. Standard UI conventions (consistent navigation bar, breadcrumb, help icon, and confirmation dialogs for destructive actions) are followed on every screen.

3.2 Software Interfaces

The back end exposes a REST API (JSON over HTTPS) consumed by the front-end client and by external integrations. The system interfaces with a relational database (PostgreSQL/MySQL) or MongoDB for persistence, and with an email service (e.g., SMTP/SendGrid) for notification delivery. Optional integrations include the Slack API (for channel notifications) and the GitHub API (for linking commits/PRs to tasks).

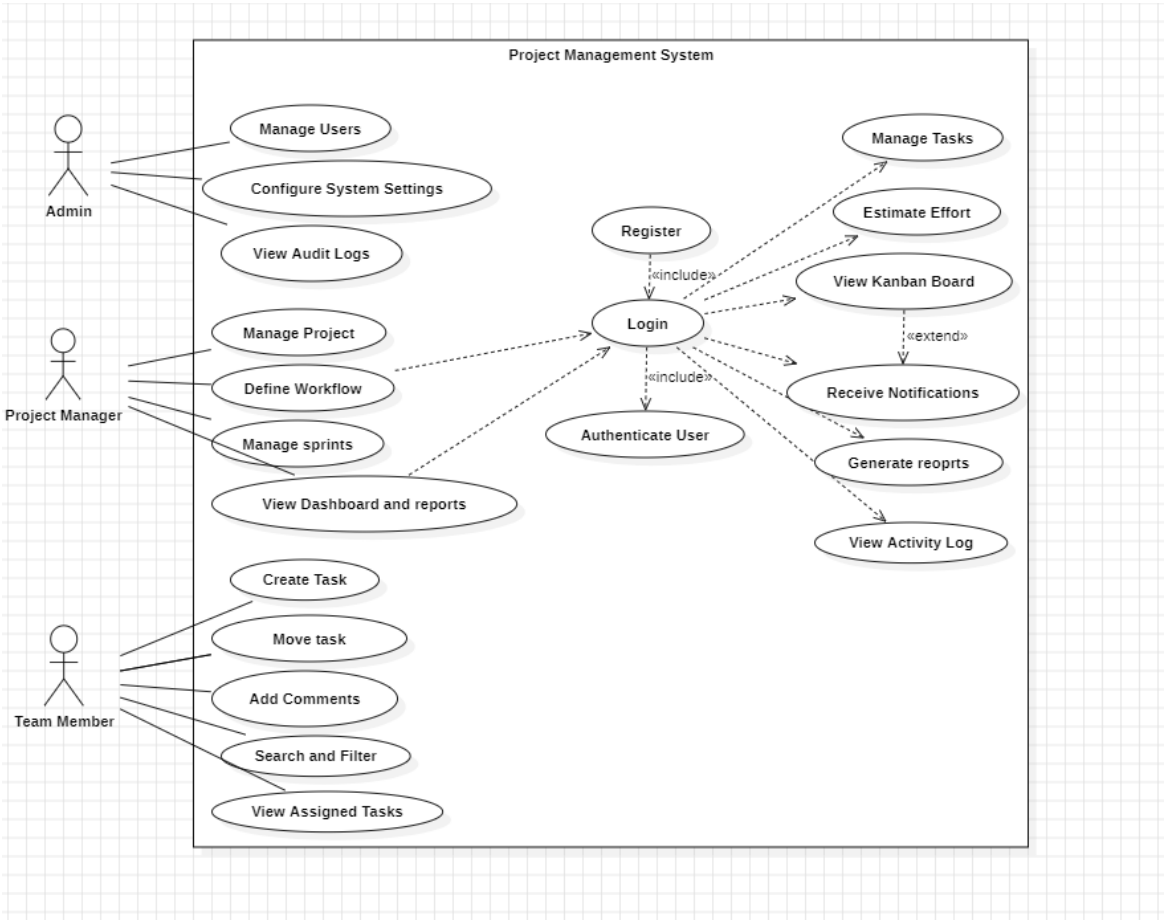
3.3 Communications Interfaces

All client-server communication uses HTTPS. Real-time or near-real-time updates (e.g., Kanban board changes, notifications) may use WebSockets or short-interval polling. Email notifications are sent via SMTP. The REST API uses standard HTTP status codes and JSON payloads, with authentication via session tokens or JWT.

4. Analysis Models

Key actors identified for the use-case model are: Admin, Project Manager, and Team Member. Primary use cases include Register/Login, Create Project, Create/Assign Task, Update Task Status (Kanban drag-and-drop), Define Workflow, Estimate Task, Plan Sprint, View Dashboard, Comment on Task, Search/Filter Tasks, Generate Report, and View Activity Log. The core entities for the entity-relationship model are User, Project, Task, Workflow, WorkflowState, Sprint, Comment, Attachment, and ActivityLog, related as: a Project has many Tasks and Sprints; a Task belongs to one Workflow State and may have many Comments and Attachments; a User can be a member of many Projects and can be assigned many Tasks.

4.1 Use Case Diagram



The Admin actor manages users, configures system settings, and views audit logs. The Project Manager manages projects, defines workflows, manages sprints, and views dashboards/reports. The Team Member creates and moves tasks, adds comments, searches/filters, and views assigned tasks. Login includes Register and Authenticate User, and in turn supports (via include/extend relationships) Manage Tasks, Estimate Effort, View Kanban Board, Receive Notifications, Generate Reports, and View Activity Log.

5. System Features

5.1 User Authentication

5.1.1 Description and Priority

Allows users to register, log in, log out, and reset passwords, with role-based access for Admin, Project Manager, and Team Member. Priority: High.

5.1.2 Stimulus/Response Sequences

- User submits registration details -> system validates and creates account, sends confirmation email.
- User submits login credentials -> system authenticates and redirects to dashboard, or shows an error.
- User requests password reset -> system emails a secure reset link -> user sets a new password.

5.1.3 Functional Requirements

- FR1.1: The system shall allow new users to register with name, email, and password.
- FR1.2: The system shall authenticate users via email/password and issue a session token.
- FR1.3: The system shall enforce role-based access control for Admin, Project Manager, and Team Member roles.
- FR1.4: The system shall allow users to reset a forgotten password via a time-limited email link.

5.2 Project Management

5.2.1 Description and Priority

Allows users to create, edit, archive, and delete projects, and to add or remove team members from a project. Priority: High.

5.2.2 Stimulus/Response Sequences

- Project Manager creates a new project with name and description -> system creates the project and assigns the creator as owner.
- Project Manager adds/removes a member -> system updates project membership and notifies the affected user.

5.2.3 Functional Requirements

- FR2.1: The system shall allow authorized users to create, edit, archive, and delete projects.
- FR2.2: The system shall allow adding or removing team members from a project.
- FR2.3: The system shall restrict project deletion/archival to Admin and Project Manager roles.

5.3 Task/Issue Management

5.3.1 Description and Priority

Allows users to create, edit, delete, and assign tasks with title, description, priority, due date, and labels. Priority: High.

5.3.2 Stimulus/Response Sequences

- User creates a task with required fields -> system saves the task and places it in the default workflow state.
- User assigns/reassigns a task -> system updates the assignee and triggers a notification.

5.3.3 Functional Requirements

- FR3.1: The system shall allow creation of tasks with title, description, priority, due date, and labels.
- FR3.2: The system shall allow editing and deleting of tasks by authorized users.
- FR3.3: The system shall allow assigning a task to exactly one team member at a time.

5.4 Kanban Board

5.4.1 Description and Priority

Visualizes tasks in columns (Backlog, To Do, In Progress, Review, Done) with drag-and-drop status updates. Priority: High.

5.4.2 Stimulus/Response Sequences

- User drags a task card from one column to another -> system updates the task's status and records the change in the activity log.

5.4.3 Functional Requirements

- FR4.1: The system shall display tasks as cards grouped into workflow-state columns.
- FR4.2: The system shall support drag-and-drop to move a task between columns.
- FR4.3: The system shall reject a drag-and-drop move that violates the project's defined workflow transitions.

5.5 Workflow Management

5.5.1 Description and Priority

Allows definition of custom workflow states and valid transitions between them for each project. Priority: Medium.

5.5.2 Stimulus/Response Sequences

- Project Manager defines workflow states and transitions -> system validates the workflow graph and saves it for the project.

5.5.3 Functional Requirements

- FR5.1: The system shall allow a Project Manager to define custom workflow states per project.
- FR5.2: The system shall allow defining valid transitions between workflow states.
- FR5.3: The system shall enforce that a task can only move along a defined transition.

5.6 Estimation

5.6.1 Description and Priority

Allows assigning story points or time estimates to tasks and tracking estimated versus actual effort. Priority: Medium.

5.6.2 Stimulus/Response Sequences

- User enters a story-point or time estimate on a task -> system stores the estimate and includes it in velocity calculations.

- User logs actual time/effort spent -> system compares it against the estimate.

5.6.3 Functional Requirements

- FR6.1: The system shall allow entry of a story-point or time estimate per task.
- FR6.2: The system shall allow logging of actual effort spent on a task.
- FR6.3: The system shall display estimated vs. actual effort per task and per sprint.

5.7 Sprint/Milestone Management

5.7.1 Description and Priority

Allows creating sprints or milestones, adding tasks to a sprint backlog, and closing sprints. Priority: Medium.

5.7.2 Stimulus/Response Sequences

- Project Manager creates a sprint with a start/end date -> system creates the sprint and an empty backlog.
- Project Manager closes a sprint -> system moves incomplete tasks to the next sprint or the backlog and finalizes velocity metrics.

5.7.3 Functional Requirements

- FR7.1: The system shall allow creation of sprints/milestones with start and end dates.
- FR7.2: The system shall allow adding and removing tasks from a sprint backlog.
- FR7.3: The system shall allow closing a sprint and shall carry over incomplete tasks.

5.8 Visibility & Dashboard

5.8.1 Description and Priority

Shows project progress, burndown/velocity charts, and task distribution by status or assignee. Priority: High.

5.8.2 Stimulus/Response Sequences

- User opens the dashboard -> system computes and renders current progress, burndown/velocity charts, and status/assignee distribution.

5.8.3 Functional Requirements

- FR8.1: The system shall display a project dashboard summarizing overall progress.
- FR8.2: The system shall render a burndown chart for the active sprint.
- FR8.3: The system shall render a velocity chart across completed sprints.
- FR8.4: The system shall show task distribution by status and by assignee.

5.9 Notifications

5.9.1 Description and Priority

Notifies users on task assignment, status changes, comments, and approaching deadlines. Priority: Medium.

5.9.2 Stimulus/Response Sequences

- A triggering event occurs (assignment, status change, comment, or approaching due date) -> system generates and delivers a notification (in-app and/or email) to the relevant user(s).

5.9.3 Functional Requirements

- FR9.1: The system shall notify a user when a task is assigned to them.
- FR9.2: The system shall notify relevant users on task status changes and new comments.
- FR9.3: The system shall send a reminder notification as a task's due date approaches.

5.10 Comments & Collaboration

5.10.1 Description and Priority

Allows users to comment on tasks and attach files or documents. Priority: Medium.

5.10.2 Stimulus/Response Sequences

- User posts a comment or uploads an attachment on a task -> system saves it, timestamps it, and notifies watchers of the task.

5.10.3 Functional Requirements

- FR10.1: The system shall allow users to add comments to a task.
- FR10.2: The system shall allow users to attach files/documents to a task.
- FR10.3: The system shall display comments in chronological order with author and timestamp.

5.11 Search & Filter

5.11.1 Description and Priority

Allows searching tasks or projects and filtering by assignee, priority, status, label, or sprint. Priority: Medium.

5.11.2 Stimulus/Response Sequences

- User enters a search term or selects filter criteria -> system returns matching tasks/projects.

5.11.3 Functional Requirements

- FR11.1: The system shall provide a keyword search across task titles and descriptions.
- FR11.2: The system shall allow filtering tasks by assignee, priority, status, label, and sprint.
- FR11.3: The system shall allow combining multiple filters in a single search.

5.12 Reporting

5.12.1 Description and Priority

Generates downloadable reports such as progress summaries, velocity, and workload. Priority: Low.

5.12.2 Stimulus/Response Sequences

- User requests a report -> system compiles the relevant data and generates a downloadable file (PDF/CSV).

5.12.3 Functional Requirements

- FR12.1: The system shall generate a project progress summary report.
- FR12.2: The system shall generate a velocity report across sprints.
- FR12.3: The system shall generate a workload report by team member.

5.13 Activity Log

5.13.1 Description and Priority

Maintains an audit trail of changes made to a task or project. Priority: Low.

5.13.2 Stimulus/Response Sequences

- Any create/update/delete action occurs on a task or project -> system records the actor, action, and timestamp in the activity log.

5.13.3 Functional Requirements

- FR13.1: The system shall log all create, update, and delete actions on tasks and projects.
- FR13.2: The system shall display a chronological activity log per task and per project.
- FR13.3: The system shall retain activity log entries for the lifetime of the project.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

Page loads and Kanban board updates shall respond within 2 seconds under normal load (NFR1).

6.2 Safety Requirements

The system does not control physical hardware and has no direct physical-safety implications. Data safety is addressed through NFR8: the system shall perform regular automated backups and support recovery in case of failure, minimizing the risk of data loss to users and projects.

6.3 Security Requirements

Passwords must be hashed, access must be role-based, and the system must use HTTPS and guard against SQL injection, XSS, and CSRF (NFR3).

6.4 Software Quality Attributes

Attribute	Requirement
Scalability (NFR2)	The system shall support a growing number of projects and users without major redesign.
Usability (NFR4)	The interface shall be intuitive, support drag-and-drop, and be responsive on desktop and tablet.
Reliability (NFR5)	The system shall maintain at least 99% uptime with graceful error handling.
Maintainability (NFR6)	The codebase shall be modular (MVC/MVT pattern), with documented APIs and unit test coverage.
Portability (NFR7)	The system shall work across major browsers and on Windows, Linux, and macOS.
Interoperability (NFR9)	The system shall expose a REST API for integration with tools such as Slack, GitHub, or email.

6.5 Business Rules

- Only Admin and Project Manager roles may create, archive, or delete a project.
- A Team Member may only edit or move tasks assigned to them or within projects they belong to.
- Only the Project Manager of a project may define or modify its workflow states and transitions.
- Domain rule: a task cannot be marked "Done" unless it currently resides in the "Review" state, enforcing a review step before closure.

7. Other Requirements

- Database requirement: the system shall use a relational database (PostgreSQL/MySQL) or MongoDB capable of supporting the data model described in Section 4.
- Legal/licensing: the project uses only open-source or freely licensed libraries and frameworks appropriate for academic use.
- Internationalization: not required for the initial release; English-only UI is acceptable.
- Reuse objective: authentication and REST API conventions may be reused across future mini-projects by the team.

Appendix A: Glossary

Term	Definition
Kanban Board	A visual board with columns representing workflow states, used to track task progress via cards.
Sprint	A fixed time period during which a defined set of tasks is planned to be completed.
Backlog	A prioritized list of tasks not yet scheduled into a sprint.
Story Point	A relative unit used to estimate the effort required to complete a task.
Burndown Chart	A chart showing remaining work in a sprint over time.
Velocity	The amount of work (story points) a team completes per sprint, averaged over time.
RBAC	Role-Based Access Control; restricting system actions based on a user's assigned role.
REST API	An architectural style for web APIs using standard HTTP methods and JSON payloads.
CSRF / XSS / SQL Injection	Common web security vulnerabilities the system must be protected against.
MVC/MVT	Model-View-Controller / Model-View-Template; architectural patterns separating data, logic, and presentation.

Appendix B: Field Layouts

Field layout for the Task entity, the primary data object created and manipulated by users:

Field	Length	Data Type	Description	Mandatory
Task ID	12	Alphanumeric	Unique identifier for the task	Y
Title	150	String	Short title of the task	Y
Description	2000	Text	Detailed description of the task	N
Priority	10	Enum	Low / Medium / High / Critical	Y
Status	20	Enum	Current workflow state (e.g., To Do, In Progress)	Y
Assignee	60	String	Name/ID of the assigned team member	N
Due Date	8	Date	Deadline for task completion	N
Story Points	3	Numeric	Effort estimate for the task	N
Labels	100	String	Comma-separated tags for categorization	N
Created Date	8	Date	Date the task was created	Y

Appendix C: Requirement Traceability Matrix

Sl. No	Requirement ID	Brief Description	Design Reference	Test Case ID
1	FR1	User Authentication	Auth Module	TC-01
2	FR3	Task/Issue Management	Task Module	TC-02
3	FR4	Kanban Board	Board UI Module	TC-03
4	FR6	Estimation	Estimation Module	TC-04
5	FR8	Visibility & Dashboard	Dashboard/Analytics Module	TC-05